HESHAM ABOUKEILA

POLYMER & MATERIALS ENGINEER

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PROFESSIONAL SUMMARY

Inventor on U.S. Patent No. 12,630,710 for enhanced polyester crystallization and developer of biobased films delivering 3x higher mechanical performance and 10x better barrier performance than commercial PBAT. Ph.D. chemical engineer specializing in polymer synthesis, processing, blown-film extrusion, structure-property relationships, surfactant formulation, and advanced characterization. Applies materials innovation to sustainable packaging and circular economy solutions, from plastics upcycling and chemical recycling to scalable product development.

TECHNICAL SKILLS

Polymer Processing: Extrusion/compounding, film blowing, compression molding

Performance Testing: Tensile testing, shear and extensional rheology, OTR/WVTR

Core Expertise: Biobased and biodegradable polymers, flexible films, crystallization, structure-property analysis

Materials Characterization: FTIR, DSC, TGA, GPC, SAXS, XRD, SEM, DMA

Surface & Colloid: Surface/interfacial tension, contact angle, bubble pressure, foam analysis

Software: Python, MATLAB, Aspen Plus, Aspen HYSYS, Symmetry

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate | University of Oklahoma | Norman, OK **Jan 2025 - Jan 2026**

- Characterized plastic-derived anionic surfactants synthesized from post-consumer recycled HDPE by measuring critical micelle concentration, surface and interfacial tension, foam stability, and calcium tolerance; demonstrated a CMC as low as 0.08 wt% and calcium tolerance of 221 ppm, competitive with conventional commercial surfactants, in a multi-institution plastics-upcycling collaboration.
- Led an industry-funded study with Church & Dwight benchmarking three commercial nonionic surfactants in potassium carbonate and sodium chloride electrolyte systems; expanded the usable HLD formulation range from approximately 4.5 to 8 and identified detergency optima of up to 90% soil removal, guiding chloride-free formulation strategies.

Graduate Research & Teaching Assistant | University of Oklahoma | Norman, OK **Aug 2021 - Dec 2024**

- Engineered and evaluated 60+ biobased biodegradable polyester formulations as alternatives to LDPE/LLDPE for flexible-film packaging, integrating polymer synthesis, crystallization, processing, and structure-property analysis.
- Demonstrated 3x higher mechanical performance and 10x better barrier performance than commercial PBAT through thermomechanical, rheological, and permeability testing.
- Produced more than 20 m of polyester and polyethylene films by lab-scale blown-film extrusion, resulting in an 80% reduction in permeability and 160% increase in tensile strength.
- Established a nucleating-agent strategy that accelerated crystallization by 56×, reducing crystallization half-time from 28 to 0.5 minutes.

Instructor | American University of the Middle East | Egaila, Kuwait **Sep 2019 - Aug 2021**

- Delivered 7 engineering courses across 4 semesters, including 4 laboratory courses and 3 problem-solving courses, serving 150+ students per semester; incorporated MATLAB and Aspen Plus into applied instruction.
- Served as course moderator and sole instructor for an additional course, developing the complete curriculum and instructional materials from scratch.
- Produced 10+ laboratory-tour and equipment-introduction videos for Unit Operations and Fluid Mechanics laboratories in collaboration with the marketing team to support remote learning.

Sales & Service Engineer | System Technique | Cairo, Egypt **Jul 2018 - Jul 2019**

- Diagnosed and resolved system and process failures across water-treatment installations, including chemical systems, filters, pumps, and softeners, minimizing client downtime and maintaining service reliability.
- Tripled the active client base from 4 to 12 accounts within one year and met the annual sales target in 6 months by building customer-specific technical presentations and value propositions that translated technical performance into economic benefits.
- Spearheaded expansion beyond water-treatment chemicals and equipment into full-cycle swimming pool services, including design, construction, operation, and maintenance, opening a new revenue vertical for the business.

Graduate Research Assistant | NTNU / University of Bologna | Norway & Italy **Mar 2016 - Mar 2018**

- Engineered Nafion/PEGDME hybrid membranes that increased dry-state CO_2 permeability by up to 36× compared with unmodified Nafion while achieving CO_2/N_2 selectivity above 30 under humid conditions.
- Controlled membrane nanostructure through solvent casting and water treatment, increasing CO_2 permeability by approximately 650% to 482 Barrer and CO_2/N_2 selectivity by 34%, surpassing the Robeson upper bound.
- Quantified humidity effects on Aquivion ionomer membranes across six gases, demonstrating more than a 10× increase in permeability without sacrificing separation selectivity for carbon-capture and natural-gas-sweetening applications.

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EDUCATION

Ph.D., Chemical Engineering | University of Oklahoma | Norman, OK — Dec 2024

M.Sc., Chemical Engineering | University of Bologna | Bologna, Italy — Mar 2018

B.Eng., Gas and Petrochemicals Engineering | Alexandria University | Alexandria, Egypt — Jul 2014

PATENT

1,5-Pentanediol-Based Polyester Compositions with Enhanced Crystallization Rates and Methods of Manufacture.
U.S. Patent No. 12,630,710 B1, 2026. Patent link

SELECTED PUBLICATIONS

- **Structure-Property Relationships in PLA-PBAT and PLA-PPeAT Blends: Toward Stronger Biobased Biodegradable Packaging Materials.** *Journal of Materials Science, 2026.* *DOI*
- **Closed-loop Recycling of Poly(pentylene adipate-co-terephthalate) via Amine-Catalyzed Methanolysis.** *ChemSusChem, 2026.* *DOI*
- **Enhanced Crystallization of Poly(pentylene adipate-co-terephthalate) via Ionomer Nucleation.** *Polymer Engineering & Science, 2025.* *DOI*
- **Film Blowing of Biobased Biodegradable Polyesters: Poly(pentylene adipate-co-terephthalate) and Poly(dodecylene furanoate).** *Macromolecular Materials and Engineering, 2025.* *DOI*
- **Biobased Poly(dodecylene furanoate) with Inherent Advantages in Performance and Circularity.** *ChemSusChem, 2025.* *DOI*
- **Upcycling of Waste Plastics into Carboxylic Acids for Biodegradable Surfactants.** *Angewandte Chemie, 2025.* *DOI*

Additional publications: Google Scholar profile

SELECTED HONORS

- Eastman Chemical Student Award in Applied Polymer Science Finalist, ACS PMSE Division (2024)
- Dissertation Excellence Award, Gallogly College of Engineering (2024)
- ANTEC Travel Award, Society of Plastics Engineers (2024)
- Judith K. Pearson Memorial Engineering Award (2023)

LEADERSHIP & PROFESSIONAL SERVICE

- President, Society of Plastics Engineers, University of Oklahoma Student Chapter (2022 - 2024)
- Chair, Academic Affairs Committee, Graduate Student Senate (2022 - 2024)
- Reviewer, Polymer Engineering & Science (2025 - Present)
- Reviewer, Journal of Polymer Engineering (2022 - Present)
- Officer, Chemical Engineering Graduate Student Society (2021 - 2024)
- Erasmus Exchange Student, Norwegian University of Science & Technology (2017 - 2018)
- Member, American Chemical Society and American Institute of Chemical Engineers

SELECTED PRESENTATIONS

- **Invited Talk, ACS Fall Meeting (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate)
- **Oral Presentation, AIChE Annual Meeting (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol Acid: Poly(pentylene adipate-co-terephthalate)
- **Poster, SPE ANTEC (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate)

LANGUAGES

Arabic (Native) | English (Fluent) | Italian (Beginner)